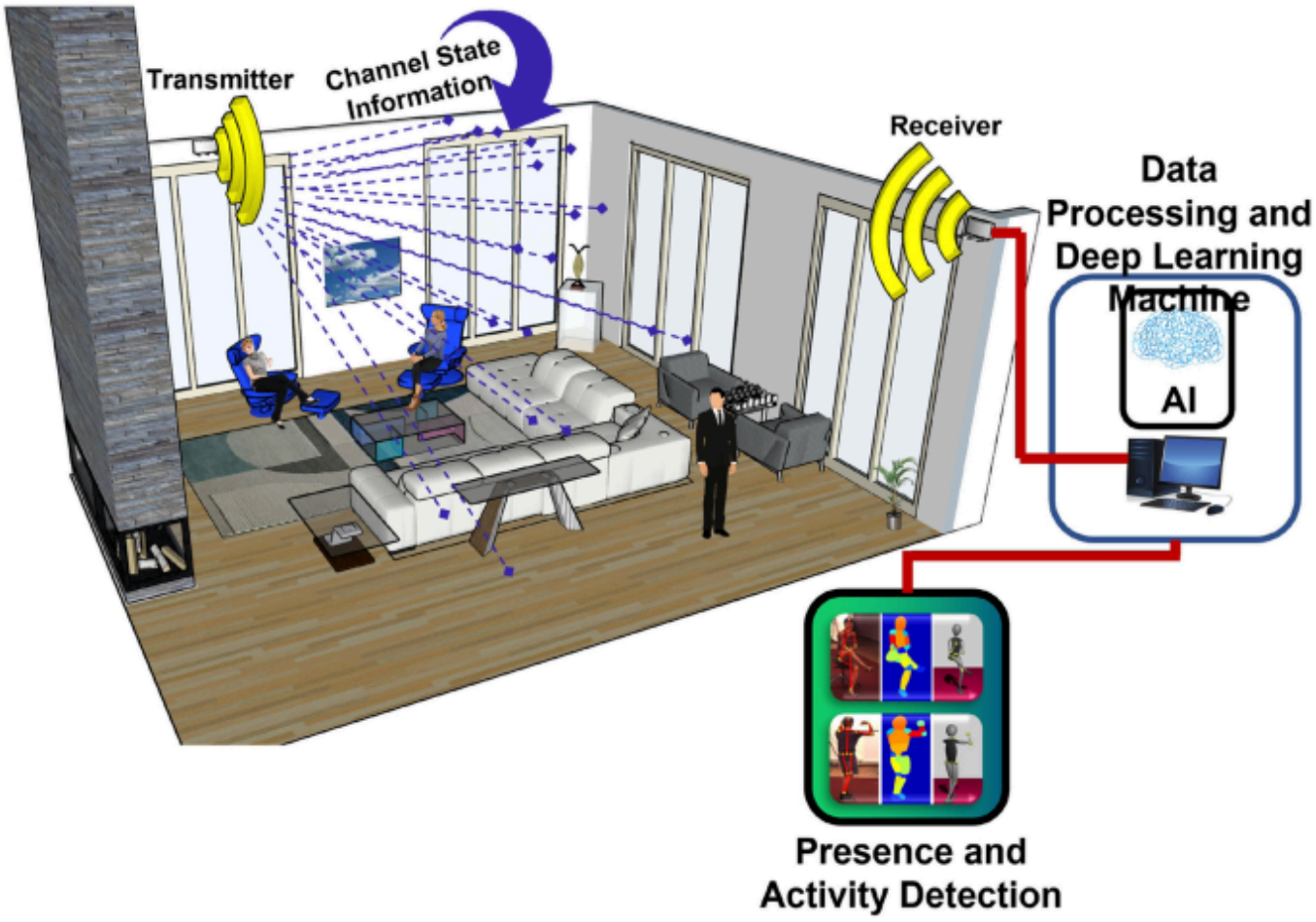
WiFi-based Human Activity Recognition (HAR)

Human Activity Recognition (HAR) via Wi-Fi Channel State Information (CSI)

- Uses standard Wi-Fi signals
- human moves between a transmitter (Router) and a receiver (Device), they scatter and reflect signals.
- **CSI** (Channel State Information) captures fine-grained signal changes (Amplitude & Phase) across multiple subcarriers (frequencies).
- Classify patterns in these signal changes into specific activities (Walking, Falling, Sitting).



WIFI HAR Dataset [Dataset Link]

- `data.csv` : Contains the raw CSI amplitude/phase for every packet received. This is the input features (X).
- `label.csv` : Contains the activity class for specific timestamps.

- **Supported Activities:** Walking, Sitting, Standing, Lying, Getting Up, Getting Down, No Activity.

- `label_boxes.csv` : Contains bounding box coordinates (x, y, w, h) of the person derived from a reference camera.
  - *Usage:* Validates if the CSI signal correlates with the person's physical location.

🧪 Dataset Characteristics

Property	Details
Activities	walking, sitting, standing, lying, getting up, getting down, no activity
# People	1
# Rooms	3
WiFi Router	TP-Link TL-WDR4300
Channel	60 (5 GHz band)
Bandwidth	40 MHz
Antennas	2 Tx × 2 Rx (MIMO)
# of Subcarriers	114
Frequency	5 GHz

XRF55 Dataset

- A Radio Frequency Dataset for Human Indoor Action Analysis.
- Developed by the **AIoT Group at Xi'an Jiaotong University** in 2024.
- 156.7GB dataset containing **42.9K samples** collected from **39 subjects**
- Data was collected across **4 distinct indoor environments** over a period of 10 months.

XRF55 uses Cross-modal alignment

- **Wi-Fi (CSI)**: 9 Wi-Fi links at 5.64GHz using 802.11n protocol (1 Tx antenna, 3 Rx antennas).
- **Dedicated mmWave Radar**: Operating at 60-64GHz to capture high-resolution motion.
- **RFID**: 23 RFID tags at 922.38MHz to track fine-grained interactions.
- **Computer Vision (Azure Kinect)**: Provides RGB, Depth, IR, and **2D human pose annotations** as a visual reference

Activities it can Detect

- **55 distinct action classes** categorized into five areas

1. **Human-Object Interactions**: Using tools or moving items.
2. **Human-Human Interactions**: Two people interacting
3. **Fitness**: Exercise and workout routines
4. **Body Motions**: General movements like walking or jumping
5. **Human-Computer Instructions**: Gestures intended to control smart devices.

Person-in-WiFi 3D: A Multi-Person 3D Pose Estimation Dataset Link

- Person-in-WiFi 3D is a Wi-Fi-based multi-person 3D pose estimation dataset. It predicts full 3D human skeletons using Wi-Fi signals
- Activities it can detect: Reaching out, Raising hands, Bending, Stretching, Sitting down, Lifting legs, Standing, Walking.
- Supports **1, 2, or 3 people at the same time**
- Instead of only detecting activities, this dataset estimates **3D body pose**, meaning the x, y, z position of each body joint.
    1. 14 body joints
    2. 3D coordinates
    3. Ground truth generated using **Azure Kinect

GaitID – Wi-Fi-based Human Gait Recognition

- GaitID is a Wi-Fi CSI dataset designed for **human gait recognition**, which means identifying people based on the way they walk
- Collected by **MobiSense Group, Tsinghua University**
- Focuses on **who is walking**, not just *what activity*
- Primary activity: **Walking (Gait)**
    1. The dataset does **not** classify many activities
    2. It focuses on:
        - Walking trajectories
        - Walking styles of different people
- Given Wi-Fi CSI → identify the person based on gait
-

Research Paper Survey

Category A : The Foundations & Hardware Accessibility

1. A Survey on Behaviour Recognition Using WiFi CSI Paper Link

- The shift from statistical methods to Deep Learning.
- Traditional Machine Learning (Random Forest, HMM) requires manual feature engineering. Deep Learning extracts features automatically.
- **LSTM (Long Short-Term Memory)** outperformed HMMs.
- LSTM achieved >90% accuracy on dynamic activities but struggled with phase errors.

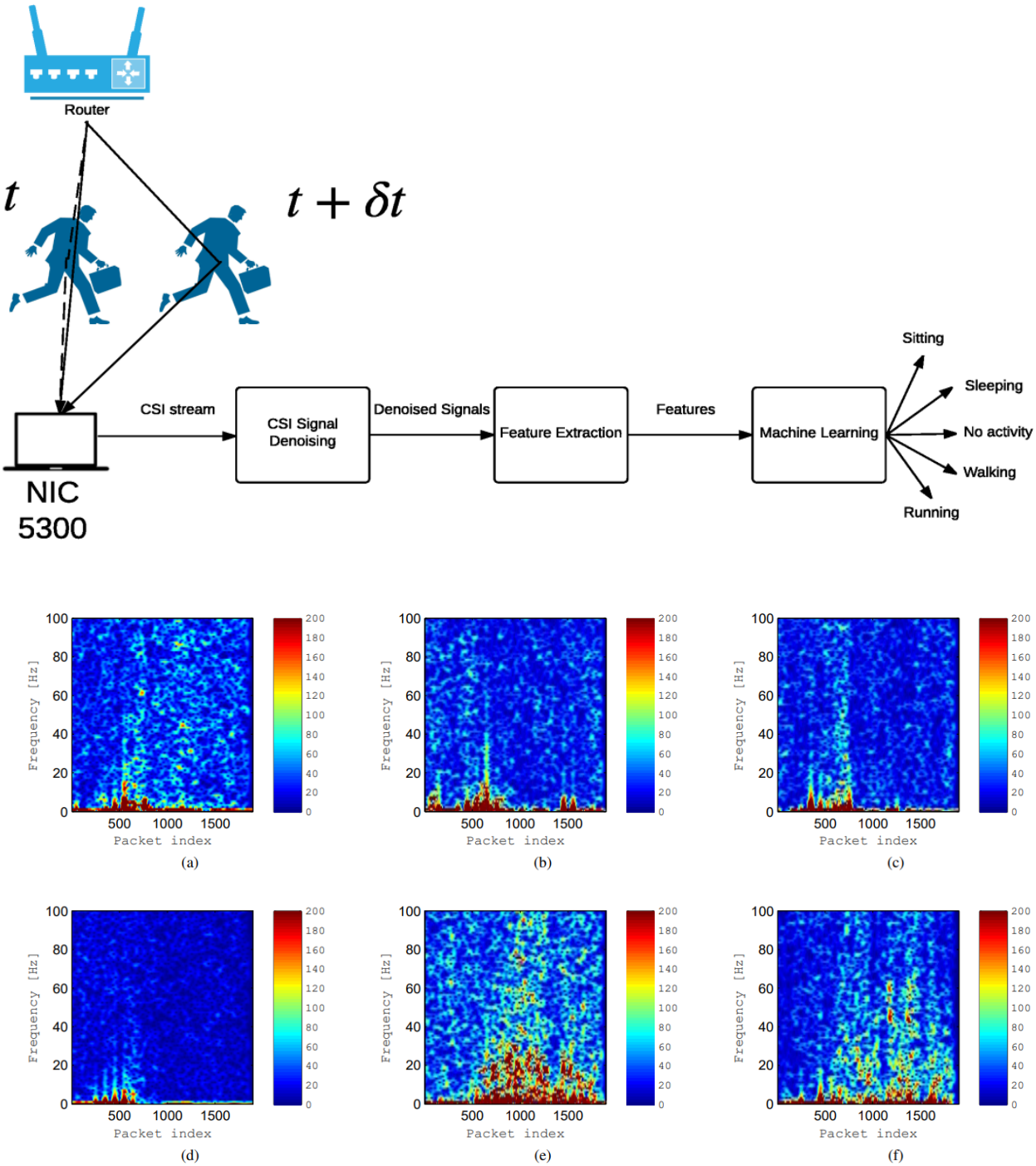
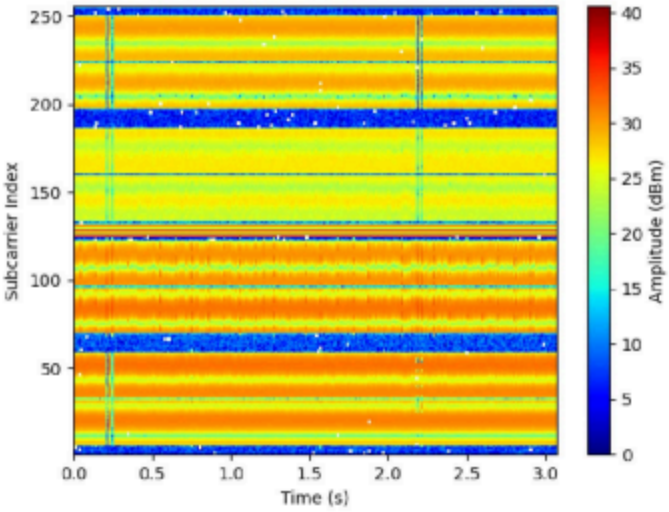


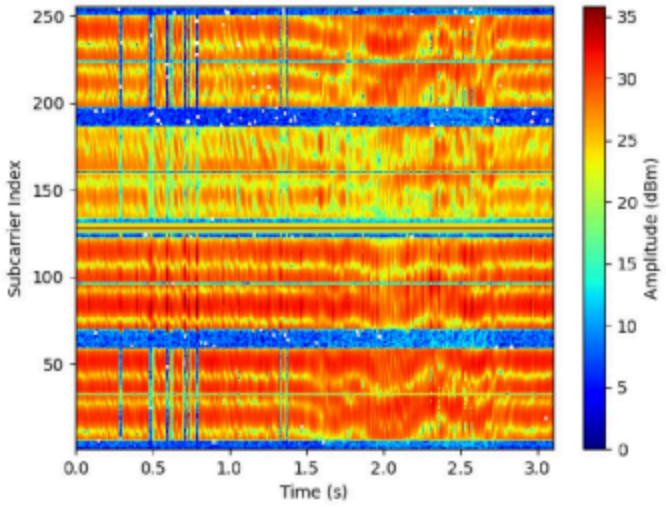
Fig. 4: The spectrogram of one subcarrier's CSI amplitude for different activities: (a) standing up, (b) sitting down, (c) laying down, (d) falling, (e) walking, (f) running.

2. WiFi-based HAR using Raspberry Pi [Paper Link]

- Democratizing hardware. Moving away from the obsolete Intel 5300 NIC to the **Raspberry Pi 4**
- 11 classes (Cooking, Brushing teeth, Petting cat , etc..)
- **DeepConvLSTM** (Convolutional layers for spatial features + LSTM for time).
- Confusion between inverse activities (Sitting Down vs. Standing Up).
- Released **CSIKit** (Python tool) for easy data parsing.



(a) still



(b) walking



3. HAR based on Wi-Fi CSI Data [Paper Link]

- End-to-end pipeline definition (Data collection -> Preprocessing -> Model).
- Comparative study of models on a "Living Lab" dataset.
- Models :
  - **InceptionTime (CNN):** Failed (Overfitting).
  - **LSTM:** ~87% Test Accuracy.
- **Weakness:** Small dataset size led to overfitting in complex models; struggled with transitional activities (Getting Up/Down).

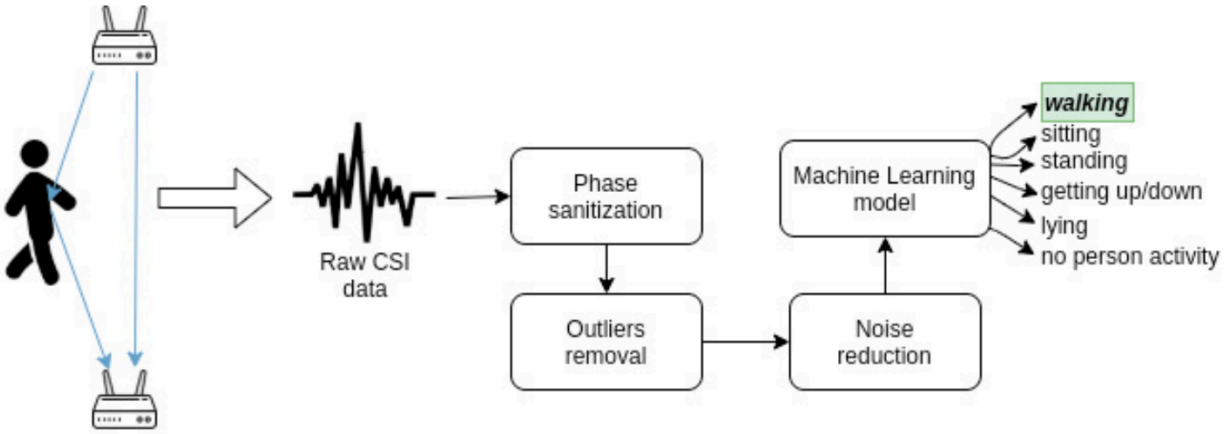


Fig. 4: System workflow from Wi-Fi router to the classification.

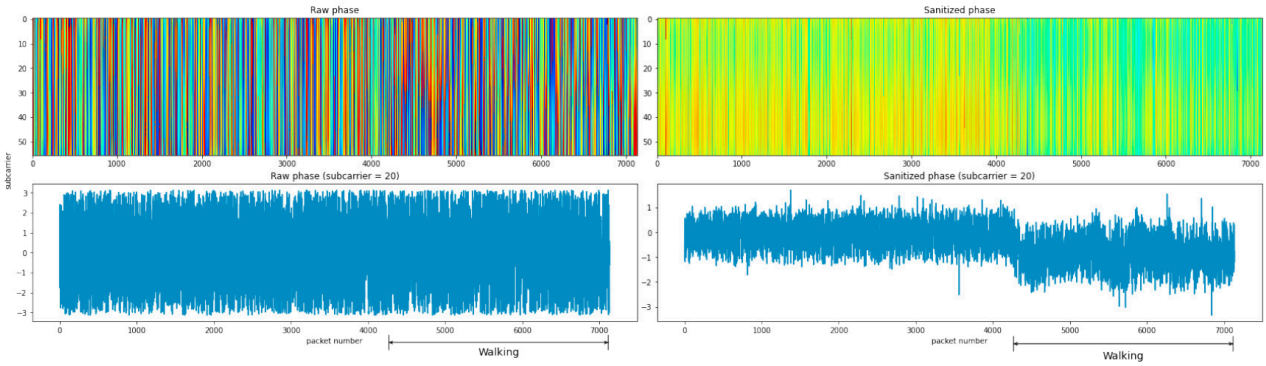
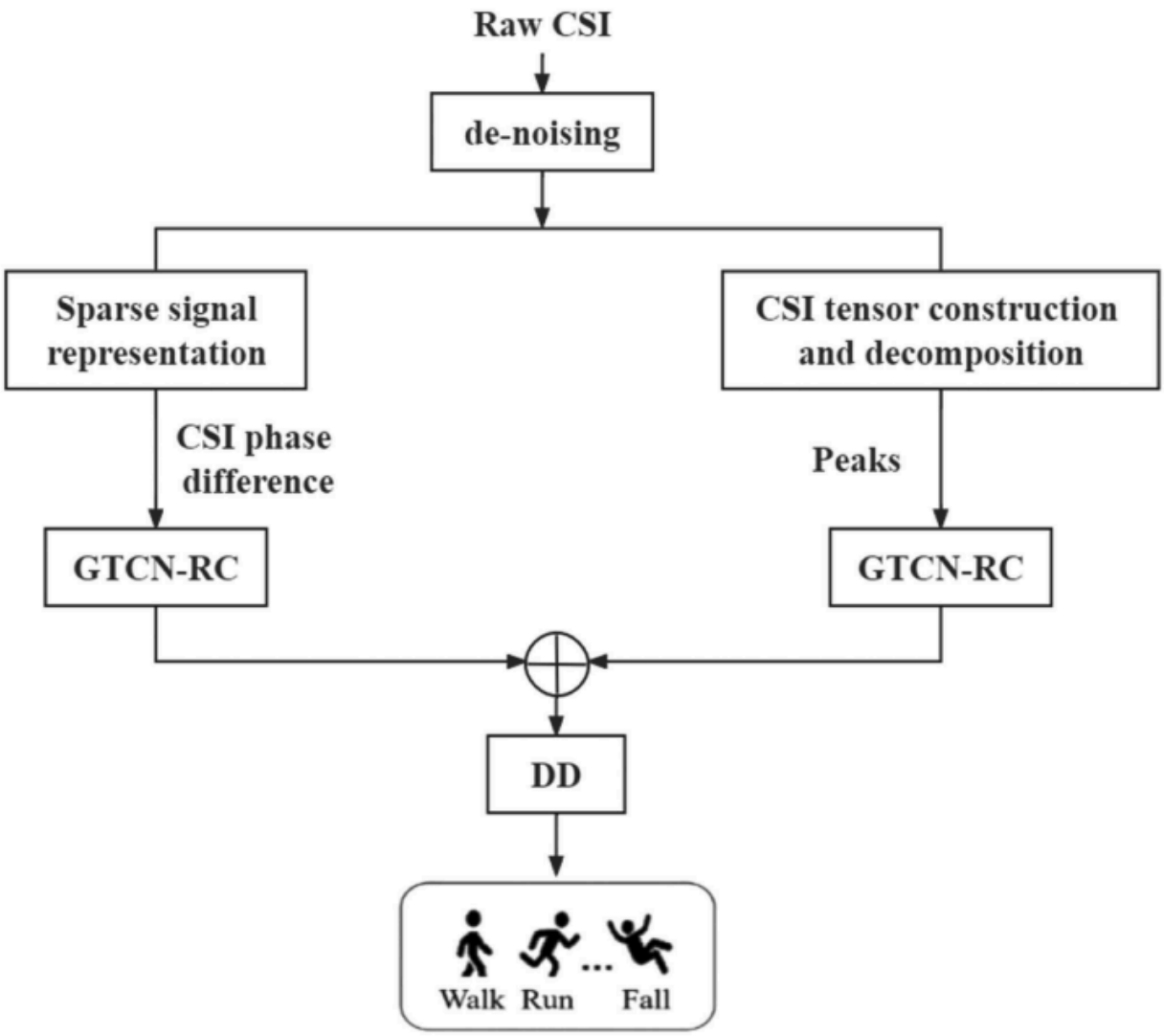


Fig. 5: Phase sanitization algorithm results: raw phase and sanitized.

Category B: Efficiency & Robustness

1. Wisor-DL: Lightweight HAR System Paper Link

    - Solving the "Cross-Domain" problem (Training in Room A, Testing in Room B) while staying fast.
    - **Reconstruction:** Uses Tensor Decomposition to clean noise.
    - **GTCN-RC** (Gated Temporal Convolutional Network) + **Dendrite Network** (replaces the heavy Dense layer).
    - accuracy drop is <1% when moving rooms (vs 15% for standard models).
    - **Weakness:** Cannot handle multi-user tracking.



Category C: Advanced Sensing & New Frontiers

1. CSI-Bench: Large-Scale In-the-Wild [Paper Link]

- **Scale:** Moving from labs to "wild" homes.
- 461 hours, 35 users, 26 environments.
- **Multi-Task Learning**. One model predicts Identity, Activity, and Proximity simultaneously.
- **ViT (Vision Transformer)** and **PatchTST**.
- Performance drops significantly on "Out of Distribution" (new) users.

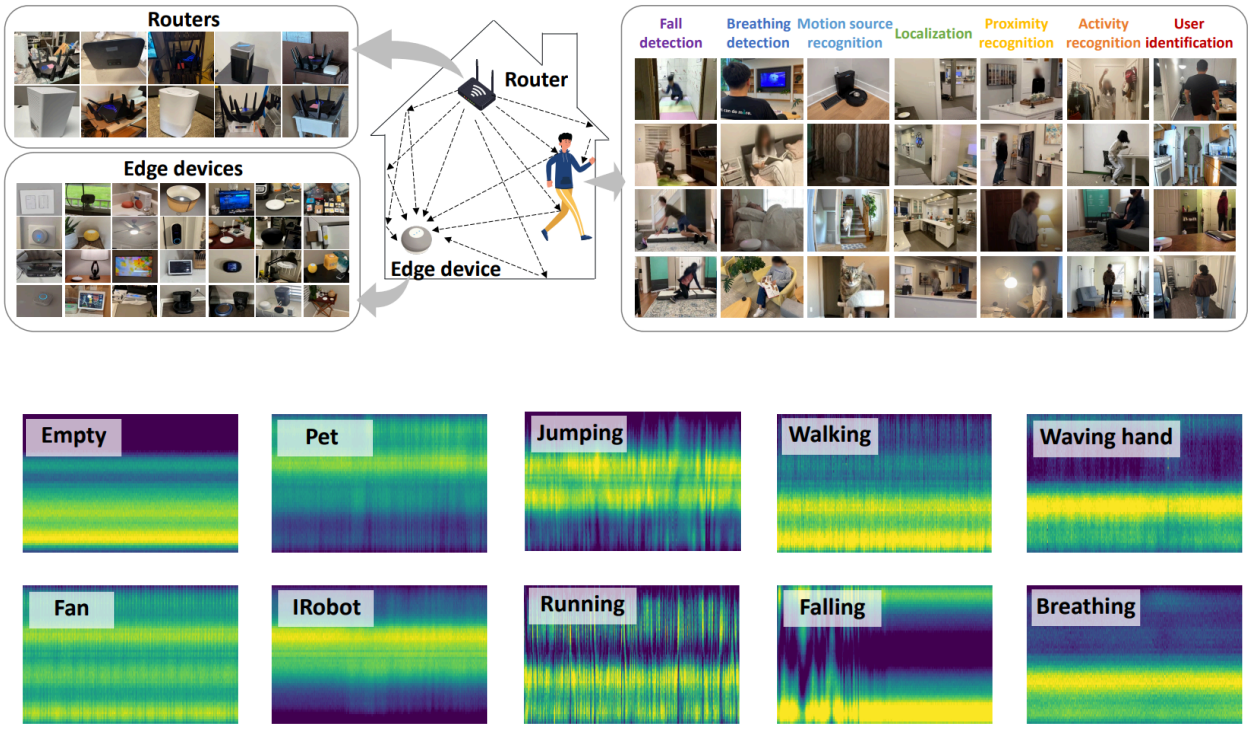
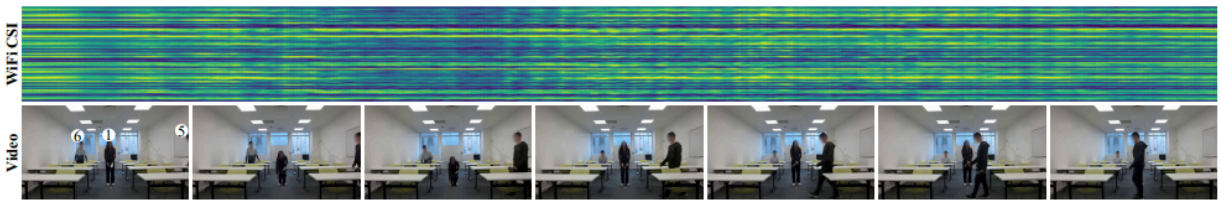


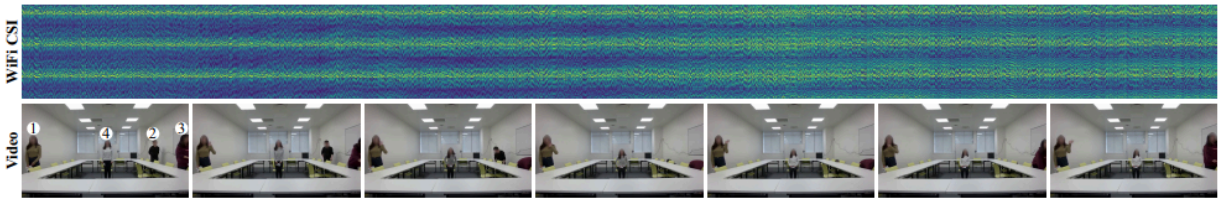
Figure 2: Representative CSI samples are shown for various scenarios, including human actions (jumping, running, walking, hand waving, falling, breathing), non-human motions (pet movement, iRobot, fan), and empty environments. In each sample, the x-axis represents time, and the y-axis represents the subcarrier index.

2. WiMANS: Multi-User Benchmark Paper Link

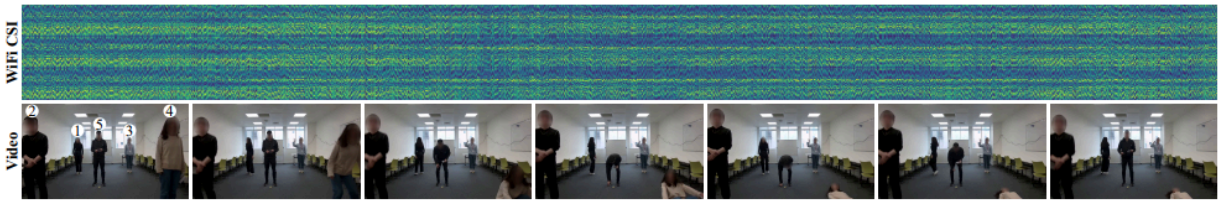
- **Multi-User Sensing.** Most datasets are single-user; this handles 0-5 concurrent users.
- Dual-band (2.4/5GHz), synchronized video.
- Benchmarked **THAT** (Two-Stream Transformer) and **CNNs**.
- **Weakness:** WiFi models struggle to distinguish *activities* when 3+ people are moving (Accuracy ~60%), even though they can identify *who* is there.



(a) User 1: Picking Up. User 5: Walking. User 6: Sitting Down.



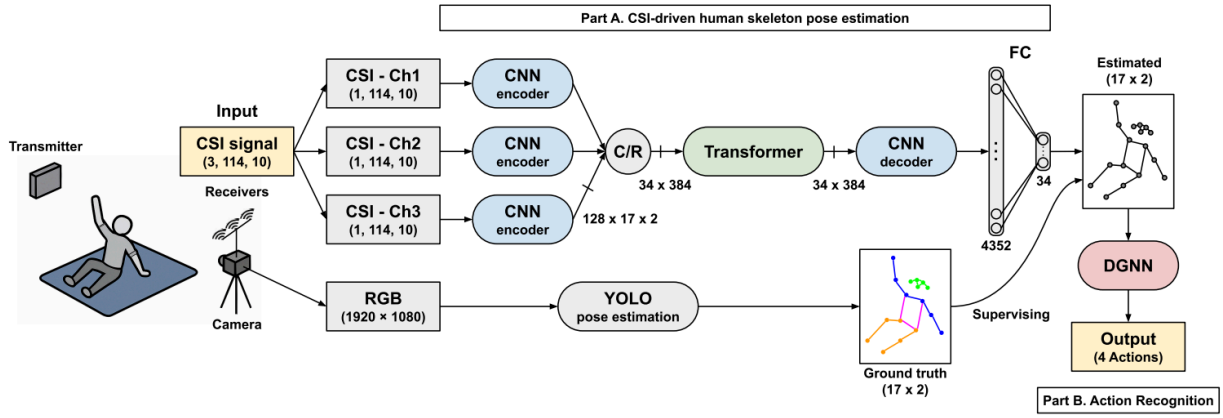
(b) User 1: Waving. User 2: Lying Down. User 3: Picking Up. User 4: Sitting Down.

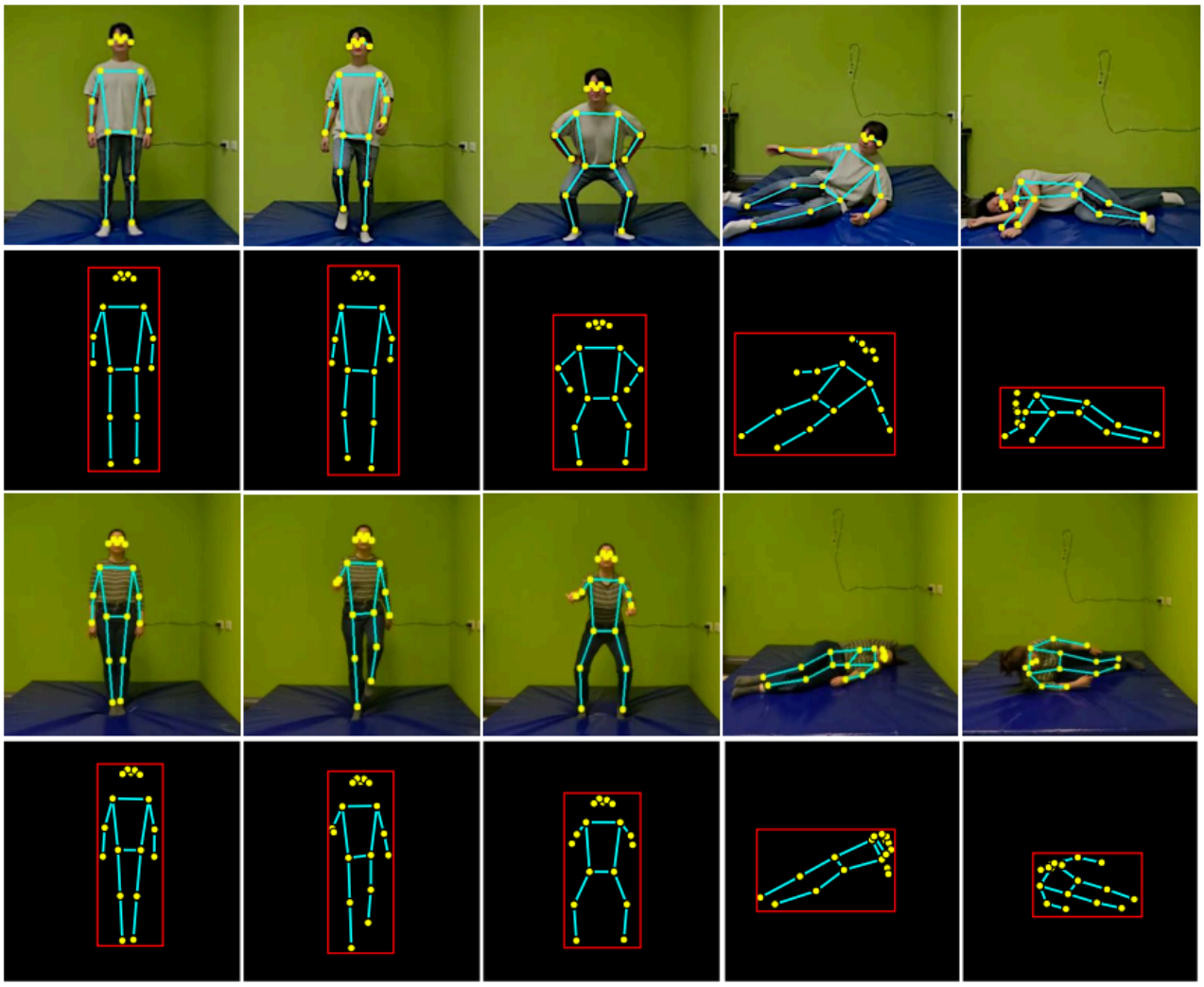


(c) User 1: Rotation. User 2: Jumping. User 3: Waving. User 4: Lying Down. User 5: Picking Up.

3. TED-Net: Fall Detection via Skeletons [Paper Link]

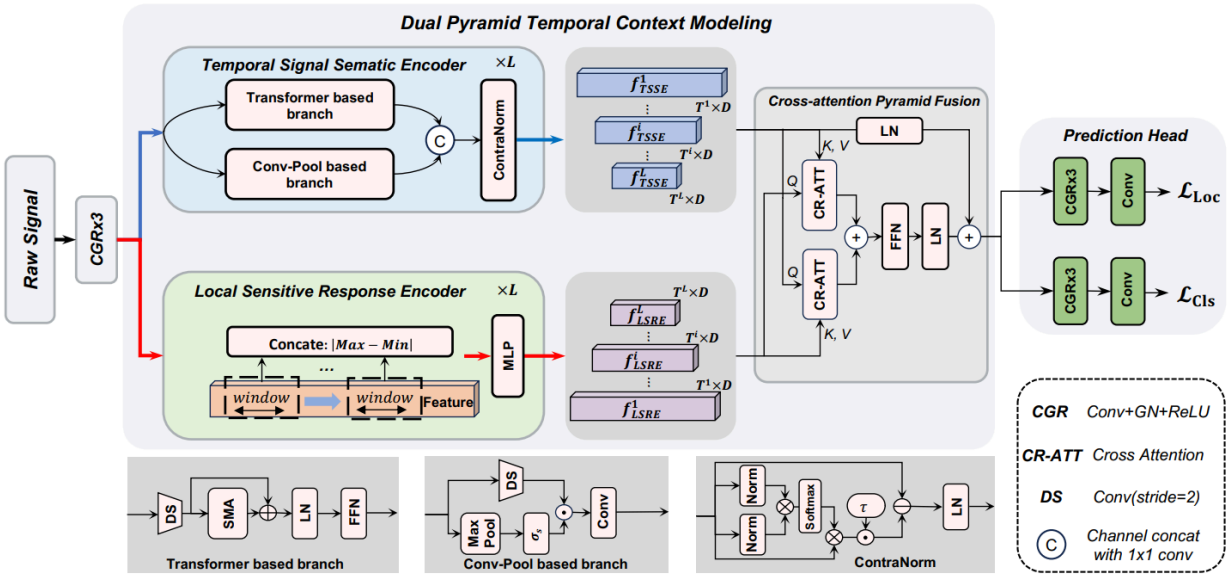
- Privacy-preserving Visualization.
- Generates a "Stick Figure" (Skeleton) from WiFi signals, then detects falls based on the skeleton's movement.
- **Transformer Encoder-Decoder** (to generate skeleton) + **DGNN** (Graph Neural Network to classify action).
- **Weakness:** Fine-grained joint accuracy (hands/feet) is lower than camera-based systems.

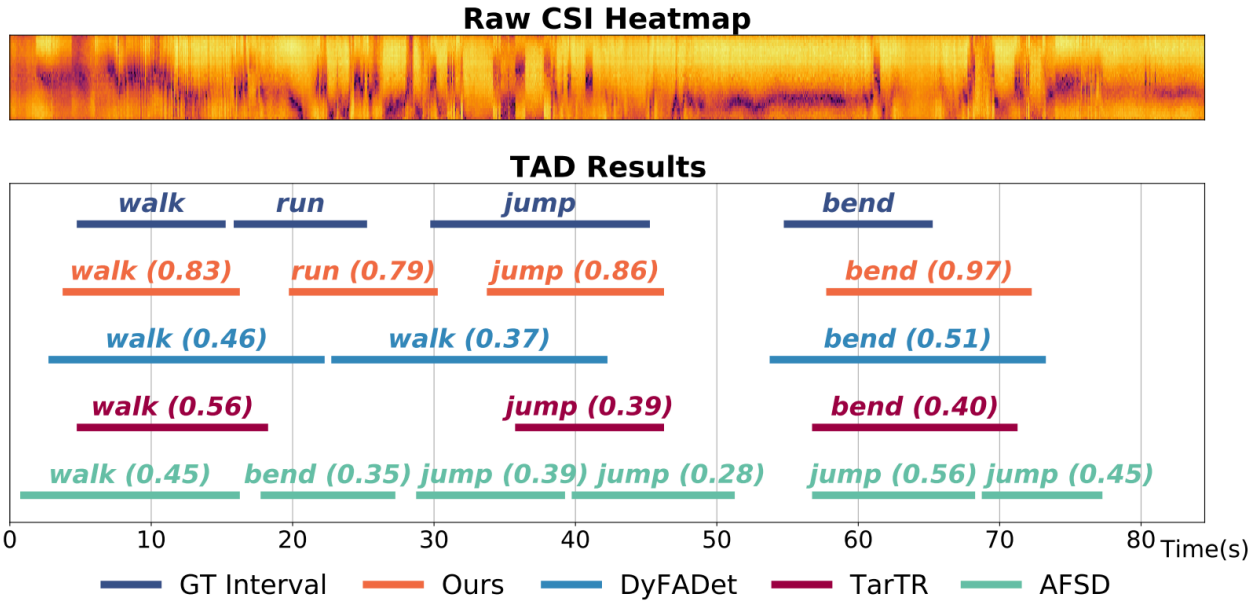




4. DPWiT: Temporal Activity Detection Paper Link

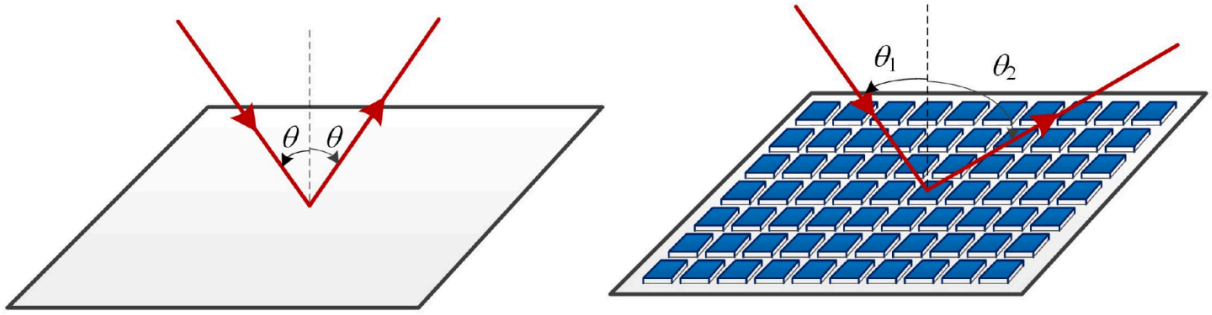
- **Continuous Stream Analysis.** Finding *when* an action starts/stops in a long recording.
- **Dual Pyramid Network**.
  - High Frequency: Finds boundaries (Start/Stop).
  - Low Frequency: Classifies the action type.
- **Weakness:** High computational complexity.





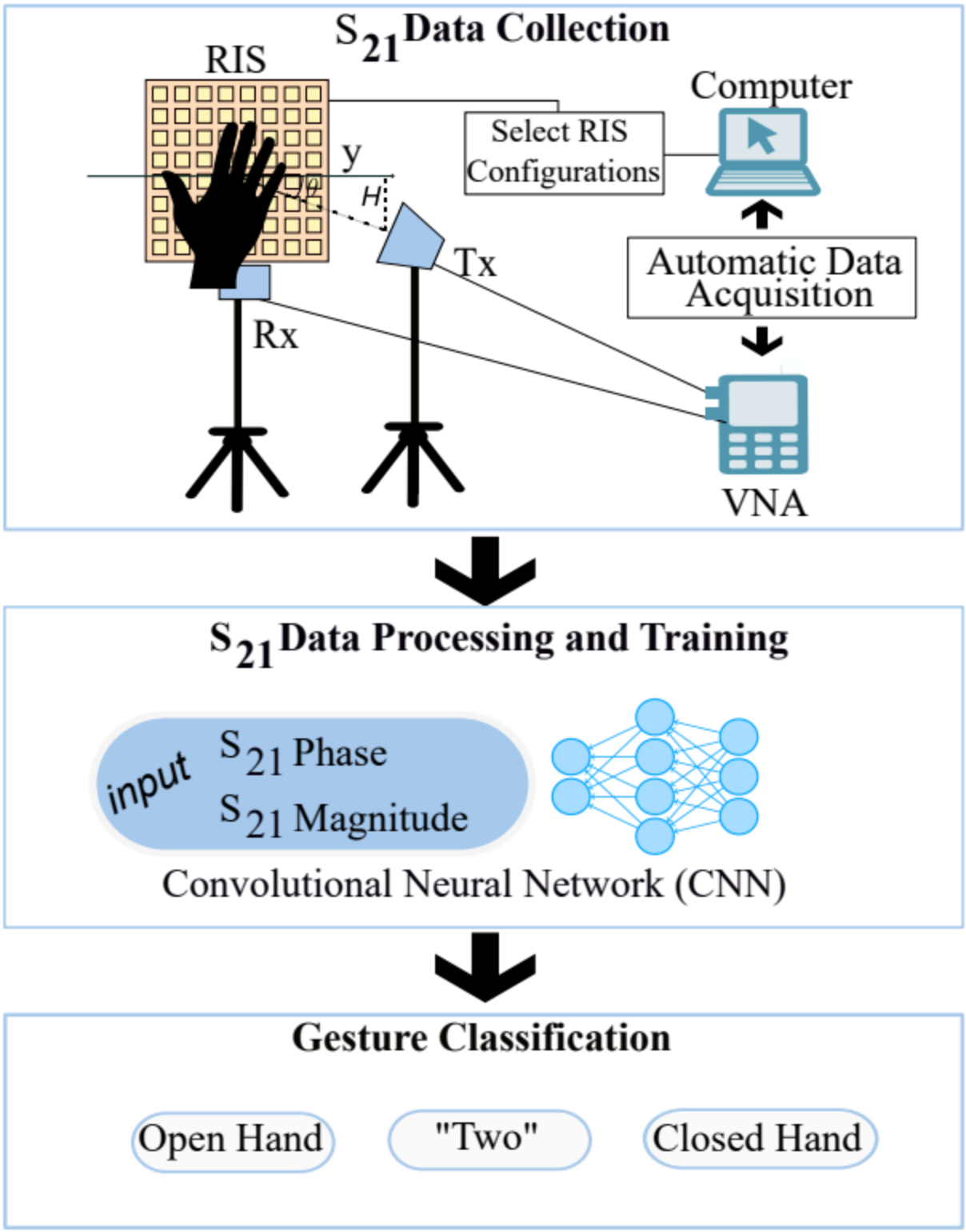
5. RIS for Wi-Fi 6E [Paper Link]

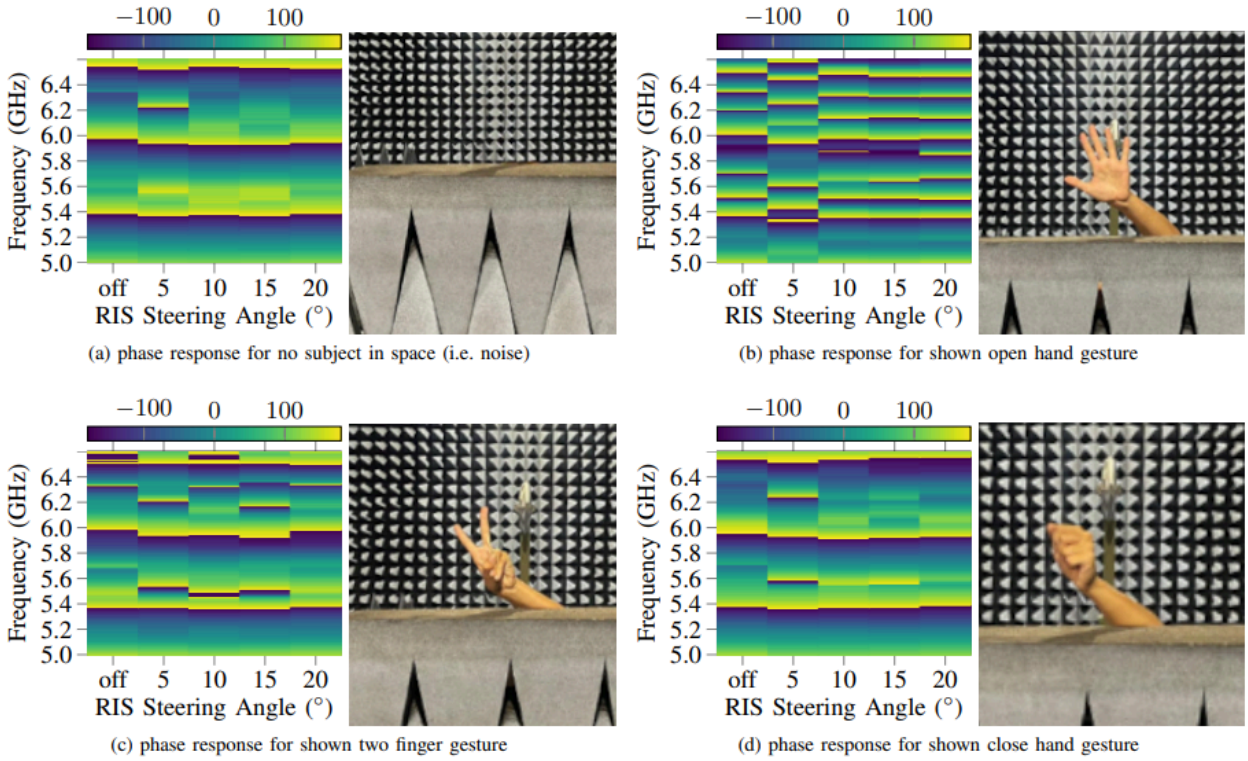
- **Hardware Augmentation**
- Uses a **Reconfigurable Intelligent Surface (RIS)** (a smart mirror) to steer 6.5GHz signals directly at the hand.
- Hand Gesture Recognition (Fine-grained).
- CNN
- **Weakness:** Requires specialized RIS hardware installation.



(a) Regular Refletor (b) RIS

Fig. 2. Regular reflector vs. RIS, which achieves non-specular reflections through programmable phase shifts (adapted from [19]).


S21 Data Collection
RIS
Computer
Select RIS Configurations
y
H
Tx
Automatic Data Acquisition
Rx
VNA
S21 Data Processing and Training
input
S21 Phase
S21 Magnitude
Convolutional Neural Network (CNN)
Gesture Classification
Open Hand
"Two"
Closed Hand



(a) phase response for no subject in space (i.e. noise)

(b) phase response for shown open hand gesture

(c) phase response for shown two finger gesture

(d) phase response for shown close hand gesture

Fig. 16. Phase responses for a frequency range of 5.0 GHz to 6.5 GHz for several RIS configurations and hand gestures.

State of the Art

1. **Models:** Transformers (ViT, TED-Net) are replacing LSTMs for complex tasks, but Lightweight CNNs (Wisor-DL) are best for edge devices.
2. **Challenges:**
   - **Single User:** Solved (98%+ accuracy).
   - **Multi-User:** Unsolved (~60% accuracy - *WiMANS*).
   - **New Environments:** Solvable via signal reconstruction (*Wisor-DL*).
3. Using "Smart Surfaces" (RIS) to steer signals for high-precision gesture recognition.